



# Phenotypic and genotypic antimicrobial resistance patterns of *Escherichia coli* and *Klebsiella* isolated from dairy farm milk, farm slurry and water in Punjab, India

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## Abstract

Antibiotic resistance is a mushrooming pandemic at national and international levels which if not controlled at this very moment, can lead to global problems. Main reason for emerging bacterial resistance is repeated exposure of bacteria to antimicrobial agents and access of bacteria to increasingly large pools of antimicrobial resistance genes in mixed bacterial populations. A total of 51 villages were sampled in the current study contributing to a total of 153 farms. A total of 612 samples comprising 153 each of raw pooled milk samples, slurry, animal drinking water and human drinking water were gathered from small, medium and large farms located in all seven tehsils of Ludhiana district of Punjab. In addition to that, 37 samples of village pond water were also collected from the targeted villages. Out of total 153 slurry, raw pooled milk samples, animal drinking water and human drinking water samples (each), the prevalence of 24.8%, 60%, 26.7% and 16.3% was found for *E. coli* respectively. On the other hand, for *Klebsiella*, the overall prevalence of 19.6%, 51%, 20.2% and 5.8% was found from slurry, raw pooled milk samples, animal drinking water and human drinking water respectively. In all matrices, the comparative frequency of resistance genes in positive isolates of *E. coli* and *K. pneumoniae* was: *tetA* > *tetB* > *tetC*, *qnrS* > *qnrB* > *qnrA*, *sulII* > *sulI* > *sulIII*. The highest proportion of resistance genes was found in slurry (193 genes) followed by milk (71 genes). The overall pattern of resistant genes was *tetA* > *sulII* > *qnrS*. In conclusion, data from the present study suggested that commensal *E. coli* and *Klebsiella* may act as reservoirs of antimicrobial drug resistance genes which may be mobilised into human populations and untreated animal waste may be considered an important source of resistant bacteria leading to environmental pollution.

**Keywords** Antibiotic resistance · Milk · Slurry · Water · *E. coli* and *K. pneumoniae* · Resistance genes

## Introduction

Antibiotic resistance is a global pandemic at national and international levels which if not controlled at this very moment, can lead to global problems. Every year drug-resistant infections lead to 58000 neonatal sepsis deaths/mortalities in India alone (Laxminarayan and Chaudhury 2013). Due to intensification in animal farming and raising demands because of expanding population antibiotic use in animals is expected to double in BRICS countries (Brazil, Russia, India, China

and South Africa) by 2030 and in 2010 India accounted for 3% of global antibiotic use in poultry, swine and cattle (Van Boeckel et al. 2015).

In relation to intensive use of antibiotics in livestock, there is also concern about the presence of pharmaceutical residues in animal manure and urine that may pass to the environment (Loh et al. 2018). The excreted antibiotics end up in soils through direct manure application or indirectly through grazing livestock (Xie et al. 2017). Sometimes, the antibiotic metabolites can transform back into their parent compound after excretion (Cycon et al. 2019). These antibiotics can have serious effects on human and environmental health as they can reach the groundwater or rinse off into surface waters (Kraemer et al. 2019).

In India, the cattle and buffalo population of India is approximately 191.2 million and 102.4 million respectively (DADH and F 2012). The average daily production of dung in cattle and buffalo is 18–30 kg and 25–40 kg respectively.

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Hence, we can estimate that every year, roughly 1095 million metric tonnes of dung is produced in India by cattle and buffalo (ICAR handbook 2013).

Analysis of antibiotics in animal faeces and urine is expected to be valuable in several ways as it may help in providing insight in the possible formation on the relation between residues and resistance and about dissemination of antibiotics to the environment and possible ecotoxicological effects (Grenni et al. 2018). In recent years, the increase in antibiotic resistance is observed in pathogens which were earlier not there. Antibiotic resistance genes (ARGs) are naturally present in the chromosomes of environmental bacteria but are now presenting in plasmids under selection pressure from antibiotics (Wright et al. 1998). The main reason for emerging bacterial resistance is repeated exposure of bacteria to antimicrobial agents and access of bacteria to increasingly large pools of antimicrobial resistance genes in mixed bacterial populations (Guerillot et al. 2018). The use of antibiotics inevitably selects both commensals and pathogenic bacteria. Human misuse of antibiotics has clearly put unnatural selective pressure on commensals and bacteria which has accelerated their evolutionary process towards resistance. The adaptation of microbes to the antibiotics increases with repeated use. Co-selection (same class) and cross selection (unrelated) to different antibiotics are exhibited by certain microbes along with direct selection (O'Brien 2002).

There is a limited documentation about the exact amount of antibiotics consumed within dairy farming in India hence it is difficult to elicit data on antibiotic consumption specific to the dairy industry (Mutua et al. 2020). There are still various critical knowledge gaps which require utmost attention. In India, environmental risk due to antibiotic residues has not been well documented as most of the studies are geographically limited to the USA and Europe (Taneja and Sharma 2019). Subtherapeutic antibiotic use in agriculture should be reduced and eventually phased out. Health professionals/veterinarians, policymakers and the public should be educated on sustainable antibiotic use (Merrett et al. 2016).

To the best of our knowledge, no such epidemiological study of antibiotic-resistant bacteria and genes in wastewaters from animal farms or in the water around the animal farms has been carried out in India and there is potential in India that the environment may play an important role in growing resistance. Thus, we believe that analysis of antibiotics in animal, faeces and urine is expected to be valuable in several ways as it may help in providing insight in the possible formation on the relation between antibiotics usage and dissemination of their residues and resistance bacteria to the environment which may result in potential ecotoxicological effects. In addition, it will help in carrying out risk assessments studies, prevention and control of antibiotic resistance to a certain level by identifying the factors to which emergence of resistance is most sensitive and thus informing future research

studies and potential interventions in India and preparing and implementing a safety statement and keeping both up to date will play a crucial role in reducing the chances of further development of AMR. Hence, keeping all these points in view, the present study was designed with the following objectives: microbiological evaluation of farm waste and surrounding surface water samples, determination of antibiotic resistance pattern and abundance of antibiotic resistance-associated genes.

## Materials and methods

### Media, chemicals and bacterial strains

Bacteriological media culture used in the present study was procured from Hi Media Laboratories (Mumbai). Chemicals and reagents such as Gotaq green, nuclease-free water and TBE buffer used were of analytical and/or molecular biological grade from Promega (Madison, Wisconsin, USA), Qiagen (Germany) and other reputed National and International firms. Primers used for the amplification of antibiotic-resistant gene were procured from Eurofins Pvt. Limited (India). *E. coli* strain ATCC 25922 and *K. pneumonia* strain ATCC 700603 were used as a control strain for antibiotic disc sensitivity test (AST).

### Site description and sample collection

Punjab is an agrarian state located in north-western India with land mass area of 50,362 km<sup>2</sup>. It is situated between 29° 30' north to 32° 32' north latitude and 73° 55' east to 76° 50' east longitudes at an altitude ranging between 180 and 300 m above sea level. Ludhiana district is divided into 7 tehsils namely, Ludhiana west, Ludhiana east, Samrala, Raikot, Payal, Khanna and Jagraon. There are a total of 906 villages in these seven tehsils. A tehsil is an administrative division in some countries of the Indian subcontinent that is usually translated to the township. It is a subdivision of the area within a district including the designated city, town, hamlet or other populated place that serves as its administrative centre, with possible additional towns, and usually a number of villages. Ludhiana west, Ludhiana east, Samrala, Raikot, Payal, Khanna and Jagraon have 132, 184, 194, 74, 111, 72 and 139 villages respectively. Ludhiana has an animal population of 765,470 (DAHD and F 2012).

The study was conducted in all seven tehsils of Ludhiana district in Northern India. The sampling was designed for a cross-sectional study. The sample size was estimated using the Epitools software (Sergeant, ESG 2019). At a given expected prevalence of 16% (Moudgil 2017), a confidence interval of 95% with a 6% desired precision gave a sampling target of 144 dairy farms. The dairy farms were classified according to

their herd size into small (less than 10 animals), medium (10–30 animals) and large (more than 30 animals) farms (Fig. 1). The categorization was based on the fact that the dairy industry of Punjab is governed by unorganized and organised sector. The unorganized farms dominating the industry are mainly owned by poorly uneducated farmers. The low level of knowledge of small farmers (Moudgil 2017) can cause a variation in antibiotic gene frequency due to poor knowledge of antibiotics, non-judicious use of antibiotics, poor knowledge of antibiotic resistance, environmental safety and importance of following good practices compared to large and medium dairy farmers. Hence, the classification-based size of operation is important especially when the study has been conducted in a developing nation like India where dairy farms are mainly owned by poor uneducated farmers. A total of 51 villages were sampled in the current study. Out of 153 dairy farms, 51 dairy farms each (small, medium and large) were visited for sampling. The study was based on a randomized design. A comprehensive list of all farms on each village from where to draw the samples in the study was available. The dairy farms from all villages were randomly selected using the EpiTools software (Sergeant, ESG 2019). A total of around 663 samples were collected from seven tehsils of Ludhiana district in Punjab. The samples consisted of milk, dairy slurry, water (animal drinking water and human drinking water) and pond water (Fig. 2). A total of 153 pooled samples of each matrix were collected from all three types of farms. In each dairy farm, two parallel samples each were taken of the targeted matrixes. All samples collected for microbiological analysis were mixed on site with an equal volume of 100% ethanol for biomass fixation. Fixed samples were kept on ice packs in ice boxes, and immediately delivered to the laboratory. At each farm, slurry was collected from 3–4 different sites; it was pooled, and then, a representative sample of approx. 100 g was collected in plastic bags. From the same

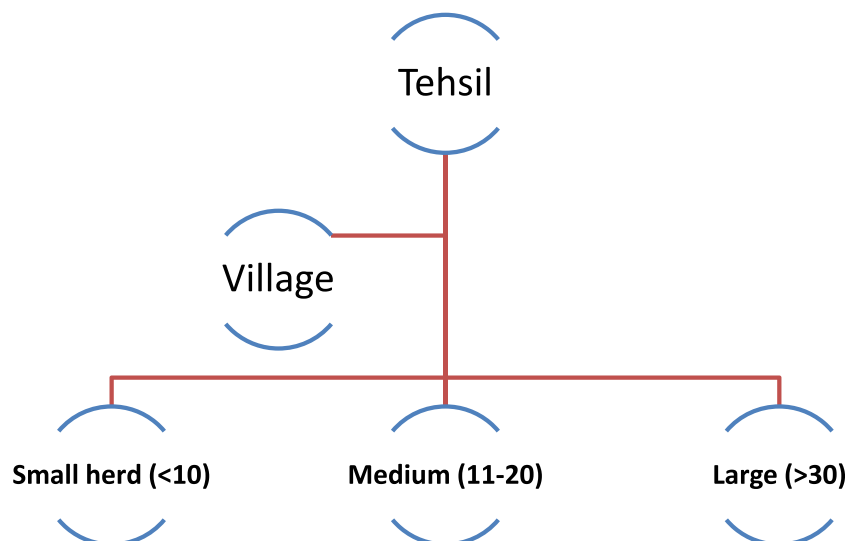
farm, 50 ml of unboiled milk sample was also collected in properly labelled sterilized tubes from the containers in which the milk of all the dairy animals was pooled together. In addition, one sample each was taken from one human and one animal drinking water container. About 100 ml of water sample was collected in sterilized plastic tubes. Pond water samples from each of the randomly selected villages were also collected. Pond water was collected from 3–4 different sites of the pond and pooled together to get approx. 100 ml of pond water. All the samples were transported to laboratory on the same day of collection under chilled conditions and stored at  $-20^{\circ}\text{C}$ . The samples were analysed within 48 h of collection. The samples were directly placed in cool boxes with ice bags ( $\sim 4^{\circ}\text{C}$ ), and remained in the field for 1–3 h (depending on the local logistics). After the sample collection was complete, the samples were transported with ice bags by car to the laboratory for further analysis. Thus, during the whole time from sample collection to its reaching the laboratory, the samples remained in cool boxes containing ice bags ( $\sim 4^{\circ}\text{C}$ ).

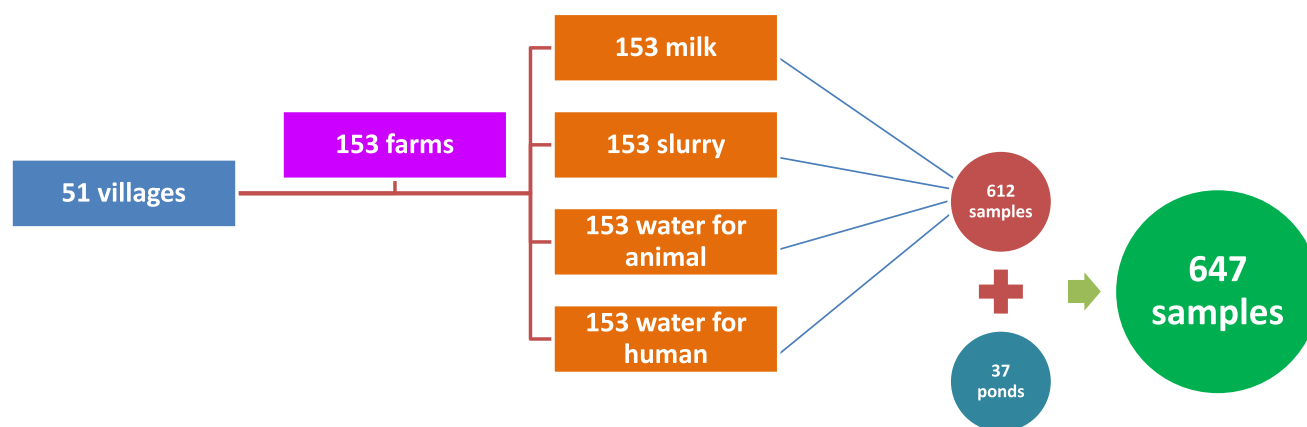
## Microbiological analysis

### Isolation of bacteria and their biochemical confirmation

From each representative sample, 1 ml water/milk/slurry were inoculated into respective tubes with 9 ml MacConkey's broth using 1 ml autoclaved tips and incubated at  $37^{\circ}\text{C}$  for 18–24 h. The enriched inoculum from broth was streaked on to Eosin-methylene blue agar for *E. coli* and on MacConkey lactose agar for *Klebsiella* and plates were incubated (Remi laboratories, India) at  $37^{\circ}\text{C}$  for 18–24 h. On EMB, *E. coli* gave a distinctive metallic green sheen and *Klebsiella* on MLA produced mucoid colonies and fermented lactose. After that, the separation of pure colonies was done on brain heart infusion agar. Isolated organisms were identified biochemically in

**Fig. 1** Sample collection plan at different levels in the study





**Fig. 2** Sample size and collection procedure of different matrices at each level in the study

systematic way following standard methods (Quinn et al. 2010). For biochemical tests, KB001 HiIMViC™ biochemical test kit (HiMedia Labs, Mumbai) was used as per manufacturer's recommendations for identification of *E. coli* and *K. pneumoniae*.

### Antibiotic susceptibility testing

All the isolates were also screened for susceptibility to antibiotics (sensidisc evaluation) namely sulphadiazine, enrofloxacin and oxytetracycline through disc diffusion method for determining phenotypic resistance pattern (Bauer et al. 1966). *E. coli* strain ATCC 25922 and *K. pneumoniae* ATCC 700603 strains were used as a control strain for AST. The zones of inhibition around the discs were measured to the nearest millimetre and interpreted as per the manufacturer cut off criteria. Using reference tables, the size of zone can be related to whether the organism is susceptible (S), intermediately susceptible (I) or resistant (R) to that antibiotic.

### Genomic DNA extraction by kit method

Genomic DNA extraction was done from *E. coli* and *K. pneumoniae* using kit (QIAamp DNA mini extraction kit, Qiagen) as per the given protocol.

### Optimization of multiplex PCR for antibiotic-resistant genes

All DNA amplification reactions were carried out in Mastercycler Gradient Thermocycler (Eppendorf, Germany) with a pre-heated lid. The PCR amplification was carried out in a total reaction volume of 25 µl containing GoTaq green master mix (Promega, USA), 10 pmol/µl of *sulI*/tetA/*qnrA* (F) and *sulI*/tetA/*qnrA* (R) primers each, 10 pmol/µl *sulII*/tetB/*qnrB* (F) and *sulII*/tetB/*qnrB* (R) primers each and 10 pmol/µl *sulIII*/tetC/*qnrS* (F) and *sulIII*/tetC/*qnrS* (R) primers each, template and sterilized nuclease-free water to make up the reaction volume.

The protocol for sulphonamide-resistant genes was standardized as described by Wu et al. (2010). Multiplex PCR protocol of tetracycline-resistant genes was optimised for three sets of primers described by Kozak et al. (2009). Screening for the quinolone-resistant genes was carried out by a multiplex PCR amplification using specific primers according to Robicsek et al. (2006).

The amplified PCR products were analysed by gel-electrophoresis (Tarsons, Kolkata) with 1.5% agarose in 1 × TBE buffer. Ethidium bromide was also added to agarose gel to a final concentration of 0.5 µg/ml when it cooled to 45–50 °C. The bands in the gel were visualized by a Gel documentation system (Syngene USA Inc. brand).

## Results and discussion

### Prevalence of *E. coli* and *K. pneumoniae* in milk, slurry and water samples belonging to different farm types in Ludhiana

A total of 612 samples from various matrices viz. slurry, raw pooled milk samples, animal drinking water and human drinking water (153 each) from different farming systems (small, medium, large) spread over different tehsils of Punjab were subjected for isolation and identification of *E. coli* and *K. pneumoniae*. In addition to that, 37 samples of village pond water were also tested for the *E. coli* and *K. pneumoniae*. Out of 153 each raw pooled milk, slurry, animal drinking water and human drinking water, the prevalence of *E. coli* was found to be 24.8%, 60.0%, 26.7% and 16.3%, respectively (Table 1).

On the basis of type of farm (large, medium and small), the maximum prevalence of *E. coli* in milk samples was reported from small farms (27.4%) followed by medium (25.5%) and then large farms (21.6%). However, in slurry samples, the maximum prevalence of *E. coli* was reported from large farms (66.6%) followed by medium farms 58.8% and then small farms with the prevalence of 52.9%. The prevalence of

**Table 1** Prevalence of in farm milk, slurry and water according to farm size

|                      | Farms                         | Milk<br>(positive isolates) | Slurry<br>(positive isolates) | Animal water<br>(positive isolates) | Human water<br>(positive isolates) |
|----------------------|-------------------------------|-----------------------------|-------------------------------|-------------------------------------|------------------------------------|
| <i>E. coli</i>       | Total farms ( <i>n</i> = 153) | 38 (24.8)                   | 91 (60.0)                     | 41 (26.7)                           | 25 (16.3)                          |
| <i>E. coli</i>       | Small farm ( <i>n</i> = 51)   | 14 (27.4)                   | 27 (52.9)                     | 11 (21.5)                           | 9 (17.6)                           |
| <i>E. coli</i>       | Medium farm ( <i>n</i> = 51)  | 13 (25.5)                   | 30 (58.8)                     | 13 (25.4)                           | 9 (17.6)                           |
| <i>E. coli</i>       | Large farm ( <i>n</i> = 51)   | 11 (21.6)                   | 34 (66.6)                     | 17 (33.3)                           | 7 (13.7)                           |
| <i>K. pneumoniae</i> | Total farms ( <i>n</i> = 153) | 30 (19.6)                   | 78 (51.0)                     | 31 (20.2)                           | 9 (5.8)                            |
| <i>K. pneumoniae</i> | Small farm ( <i>n</i> = 51)   | 10 (19.6)                   | 22 (43.1)                     | 7 (13.7)                            | 3 (5.8)                            |
| <i>K. pneumoniae</i> | Medium farm ( <i>n</i> = 51)  | 12 (23.5)                   | 25 (49.0)                     | 10 (19.6)                           | 2 (3.9)                            |
| <i>K. pneumoniae</i> | Large farm ( <i>n</i> = 51)   | 8 (15.7)                    | 31 (60.7)                     | 14 (27.4)                           | 4 (7.8)                            |

Figures in parenthesis indicate percentage positive

*E. coli* from animal water from small, medium and large farm was 21.5%, 25.4% and 33.3%, respectively, while the prevalence of *E. coli* from human drinking water samples collected from small and medium farm was 17.6%, and from large farm, it was 13.7% (Table 1).

Out of each 153 raw pooled milk, slurry, animal drinking water and human drinking water, the prevalence of *K. pneumoniae* was found to be 19.6%, 51.0%, 20.2% and 5.8%, respectively (Table 1). On the basis of type of farm (large, medium and small), maximum prevalence of *K. pneumoniae* in milk samples was reported from small farms (19.6%) followed by medium (23.5%) and then large farms (15.7%). However, in slurry samples, the maximum prevalence of *K. pneumoniae* was reported from large farms (60.7%) followed by medium farms (49.0%) and then small farms with the prevalence of 51.0%. The prevalence of *K. pneumoniae* from animal water from small, medium and large farm was 7.0%, 19.6% and 27.4%, respectively. The prevalence of *K. pneumoniae* from human drinking water samples collected from small and medium farm was 5.8% and 3.9%, respectively, while from large farm, it was 7.8%.

From 37 collected pond water, the prevalence of *E. coli* and *K. pneumoniae* was 94.5% (35/37) and 89.1% (33/37) respectively.

In Ahmedabad (India), a study conducted on drinking water samples, water from the drainage pipes and water from the leaking pipes of General Hospital from the civil hospital, 22 samples (44%) showed contamination with faecal coliform. Of the 22 positive samples, *E. coli* was isolated from 12 water samples; *K. pneumoniae* was isolated from 10 samples (Sood et al. 2015). In a study of Chennai, India, out of 567 samples of raw milk tested, 34 (5.9%) *E. coli* were isolated (Annal et al. 2018). In a similar study (Kumar and Prasad 2010) of Pantnagar, India, a study revealed that out of 135 samples, 11 samples were found contaminated with *E. coli* (8.1%). A study of three different states of India viz. West Bengal, Jharkhand and Mizoram reported 23 (17.8%) *Klebsiella pneumoniae*,

isolated from milk samples (*n* = 340) of cows (Koovapra et al. 2016). In West Bengal, Inida, *Klebsiella pneumoniae* were found in 26.41% (112/424) of the milk samples (Bandyopadhyay et al. 2018). The present study is in compliance with other studies of India. In a study (Sukhon 2003), a total of 346 *K. pneumoniae* isolates were isolated from milk and milk products out of which *Klebsiella pneumoniae*, *Klebsiella oxytoca* and *Klebsiella rhinoscleromatis* were identified in 63 (9.7%), 31 (4.7%) and 28 (4.3%) isolates, respectively. *E. coli* were isolated in 80% samples of cow dung in Assam, India (Borah et al. 2014). Fifty-seven isolates of *E. coli* were recovered from 125 drinking water sources in Jammu, India (Wani et al. 2014).

#### Antibiotic sensitivity pattern of *E. coli* and *K. pneumoniae* in milk, slurry, animal drinking water, human drinking water (farm-wise) and pond water

All the *E. coli* (*n* = 195) isolates, 38 from various milk samples, 91 from slurry samples, 41 from animal water and 25 from human drinking water were further subjected to antibiotic sensitivity pattern by disc diffusion method. Amongst *E. coli* isolates (195), slurry samples showed maximum resistance to all three antibiotics discs tested (32.9%) and least for isolates obtained from human drinking water (0%) (Table 2). Most of the isolates from slurry (91) of all farms (small, medium, large) were resistant to oxytetracycline (59.3%) followed by enrofloxacin (56%) and least for sulphadiazine (23.5%). While maximum resistance was reported from isolates obtained from slurry samples of large farms, out of 34 isolates, 64.7 % were resistant to oxytetracycline, while 61.8% and 55.9% isolates were resistant to enrofloxacin and sulphadiazine, respectively (Table 2). In milk samples, the highest resistance was found for oxytetracycline (61.5%) followed by enrofloxacin (55.7%) and sulphadiazine (42.3%).

Out of 156 *K. pneumoniae* isolates, 30 from pooled raw milk samples, 78 from slurry samples, 31 from animal water



**Table 2** Dairy farm-wise comparative resistance pattern of *E. coli* based on disc diffusion method

| Farm   | Matrix       | <i>E. coli</i> isolates | Resistance pattern based on total samples |                              |                             |                                | Resistance pattern based on positive isolates samples |                             |                                |                              |
|--------|--------------|-------------------------|-------------------------------------------|------------------------------|-----------------------------|--------------------------------|-------------------------------------------------------|-----------------------------|--------------------------------|------------------------------|
|        |              |                         | Oxytetracycline Resistance (%)            | Sulphadiazine Resistance (%) | Enrofloxacin Resistance (%) | Oxytetracycline Resistance (%) | Sulphadiazine Resistance (%)                          | Enrofloxacin Resistance (%) | Oxytetracycline Resistance (%) | Sulphadiazine Resistance (%) |
| Small  | Milk         | 14                      | 13.7                                      | 11.7                         | 13.7                        | 50.0                           | 42.8                                                  | 50.0                        | 50.0                           | 42.8                         |
|        | Slurry       | 27                      | 27.5                                      | 17.6                         | 25.5                        | 51.9                           | 33.3                                                  | 48.1                        | 51.9                           | 33.3                         |
|        | Animal water | 11                      | 7.8                                       | 3.9                          | 5.8                         | 36.3                           | 18.1                                                  | 27.2                        | 36.3                           | 18.1                         |
| Medium | Human water  | 9                       | -                                         | -                            | -                           | -                              | -                                                     | -                           | -                              | -                            |
|        | Milk         | 13                      | 9.8                                       | 7.8                          | 15.6                        | 38.5                           | 30.7                                                  | 61.5                        | 38.5                           | 30.7                         |
|        | Slurry       | 30                      | 35.3                                      | 15.7                         | 33.3                        | 60.0                           | 26.7                                                  | 56.7                        | 60.0                           | 26.7                         |
| Large  | Animal water | 13                      | 17.6                                      | 5.8                          | 11.7                        | 69.2                           | 23.0                                                  | 46.1                        | 69.2                           | 23.0                         |
|        | Human water  | 9                       | -                                         | -                            | -                           | -                              | -                                                     | -0                          | -                              | -                            |
|        | Milk         | 11                      | 9.8                                       | 3.9                          | 9.8                         | 45.5                           | 18.2                                                  | 45.5                        | 45.5                           | 18.2                         |
|        | Slurry       | 34                      | 43.1                                      | 37.2                         | 41.2                        | 64.7                           | 55.9                                                  | 61.8                        | 64.7                           | 55.9                         |
|        | Animal water | 17                      | 19.6                                      | 11.7                         | 13.7                        | 58.8                           | 35.2                                                  | 41.1                        | 58.8                           | 35.2                         |
|        | Human water  | 7                       | -                                         | -                            | -                           | -                              | -                                                     | -                           | -                              | -                            |
|        | Pond water   | 35                      | 64.8                                      | 59.4                         | 67.5                        | 68.5                           | 62.8                                                  | 71.4                        | 68.5                           | 62.8                         |

and 9 from human drinking water were further subjected to antibiotic sensitivity pattern by disc diffusion method. Amongst *K. pneumoniae* isolates (156) slurry samples showed maximum resistance to all three antibiotics tested (29.4%) and least for isolates obtained from human drinking water (0%). Most of the isolates from slurry (78) of all farms (small, medium, large) were resistant to oxytetracycline (35.8%) followed by enrofloxacin (33.3%) and least for sulphadiazine (21.7%) (Table 3), while resistance against these three antibiotics for *K. pneumoniae* isolates obtained from large farms was the highest followed by medium and least for small farm, while in milk samples, resistance pattern at farm level was the same as for slurry samples i.e. large farms> medium farms> small farms (Table 3).

While out of 37 pond water samples, 35 *E. coli* isolates showed maximum resistant genes for enrofloxacin (67.5%) followed by oxytetracycline (64.8%) and sulphadiazine (59.4%) (Table 2). *K. pneumoniae* isolates (33/37) showed maximum resistance genes for oxytetracycline (78.3%) followed by enrofloxacin (72.9%) and least for sulphadiazine (67.5%) (Table 3).

From large farm slurry samples out of 34 *E. coli* isolates, 16 were resistant to all three antibiotics (47.0%), while 7 (23.3%) slurry isolates out of 30 were multi-drug resistant in medium farms and out of total 27 small farm slurry isolates, 7 (25.9%) were found to be multi-drug resistant (Table 4). From milk, reverse trend was observed as of slurry, maximum isolates resistant to all three antibiotics was observed from small farm (28.6%) followed by middle farm (23.1%) and least from large farm (9.1%), while all *E. coli* isolates from human drinking water collected from different farms (small, medium and large) were sensitive to all antibiotics tested.

Out of 31 *K. pneumoniae*, isolates from slurry samples from large farm 13 (41.9%) were resistant to all three antibiotics while from the medium farm, 7 (28%) were resistant to all three antibiotics and 7 (31.8%) isolates from 22 were multi-drug resistant from small farm (Table 5). Nine isolates of *K. pneumoniae* from 51 total isolates from all drinking water were susceptible to test three antibiotics.

In pond water samples, the antibiogram profile indicated that for *E. coli* out of 35 isolates, 19 (54.2%) were multi-drug resistant; 11 (31.4%) were resistant for two antibiotics tested; five isolates (14.2%) were resistant to one antibiotic and no isolate was sensitive for any of antibiotic tested (Table 4). Similarly, for *K. pneumoniae* out of 33 isolates, 21 (63.6%) were multi-drug resistant; 10 (30.3%) were resistant for two antibiotics tested while 2 isolates were (6%) resistant to one antibiotic (Table 5).

On comparison with our study, Srinivasan et al. (2007) isolated 129 *E. coli* isolates from dairy cow milk and subjected them to disc sensitivity; all *E. coli* were resistant to two or more antimicrobials. 98.4% *E. coli* were resistant to ampicillin; 40.3% were resistant to streptomycin, 34.1% to

**Table 3** Dairy farm-wise comparative resistance pattern of *K. pneumoniae* based on disc diffusion

| Farm   | Matrix       | <i>K. pneumoniae</i> isolates | Resistance pattern based on total samples |                              |                             |  | Resistance pattern based on positive isolates samples |                              |                             |  |
|--------|--------------|-------------------------------|-------------------------------------------|------------------------------|-----------------------------|--|-------------------------------------------------------|------------------------------|-----------------------------|--|
|        |              |                               | Oxytetracycline Resistance (%)            | Sulphadiazine Resistance (%) | Enrofloxacin Resistance (%) |  | Oxytetracycline Resistance (%)                        | Sulphadiazine Resistance (%) | Enrofloxacin Resistance (%) |  |
| Small  | Milk         | 10                            | 5.9                                       | 1.96                         | 5.9                         |  | 30.0                                                  | 10.0                         | 30.0                        |  |
|        | Slurry       | 22                            | 23.5                                      | 17.6                         | 1.96                        |  | 54.5                                                  | 40.9                         | 4.5                         |  |
|        | Animal water | 7                             | 5.8                                       | 1.9                          | 7.8                         |  | 42.8                                                  | 14.2                         | 57.1                        |  |
|        | Human water  | 3                             | -                                         | -                            | -                           |  | -                                                     | -                            | -                           |  |
| Medium | Milk         | 12                            | 9.8                                       | 1.96                         | 11.8                        |  | 41.7                                                  | 8.3                          | 50.0                        |  |
|        | Slurry       | 25                            | 29.4                                      | 23.5                         | 27.5                        |  | 60.0                                                  | 48.0                         | 56.0                        |  |
|        | Animal water | 10                            | 11.7                                      | 5.8                          | 7.8                         |  | 60.0                                                  | 30.0                         | 40.0                        |  |
|        | Human water  | 2                             | -                                         | -                            | -                           |  | -                                                     | -                            | -                           |  |
| Large  | Milk         | 8                             | 3.9                                       | 3.9                          | 9.8                         |  | 25.0                                                  | 25.0                         | 62.5                        |  |
|        | Slurry       | 31                            | 45.1                                      | 35.3                         | 37.3                        |  | 74.2                                                  | 58.1                         | 61.3                        |  |
|        | Animal water | 14                            | 17.6                                      | 11.7                         | 17.6                        |  | 64.2                                                  | 42.8                         | 64.2                        |  |
|        | Human water  | 4                             | -                                         | -                            | -                           |  | -                                                     | -                            | -                           |  |
| -      | Pond water   | 33                            | 78.3                                      | 67.5                         | 72.9                        |  | 82.8                                                  | 71.4                         | 77.1                        |  |

**Table 4** Dairy farm-wise distribution of resistance against antibiotics in *E. coli* based on disc diffusion method

| Matrix       | Farm   | <i>E. coli</i> isolates | MDR (resistant to all antibiotics) (no of farms) | Resistant to two antibiotics (no of farms) | Resistant to one antibiotic (no of farms) | Sensitive to all antibiotics (no of farms) |
|--------------|--------|-------------------------|--------------------------------------------------|--------------------------------------------|-------------------------------------------|--------------------------------------------|
| Milk         | Small  | 14                      | 4                                                | 3                                          | 5                                         | 2                                          |
|              | Medium | 13                      | 3                                                | 3                                          | 5                                         | 2                                          |
|              | Large  | 11                      | 1                                                | 3                                          | 3                                         | 4                                          |
| Slurry       | Small  | 27                      | 7                                                | 5                                          | 7                                         | 8                                          |
|              | Medium | 30                      | 7                                                | 13                                         | 3                                         | 7                                          |
|              | Large  | 34                      | 16                                               | 6                                          | 5                                         | 7                                          |
| Animal water | Small  | 11                      | -                                                | 3                                          | 3                                         | 5                                          |
|              | Medium | 13                      | 2                                                | 5                                          | 3                                         | 3                                          |
|              | Large  | 17                      | 1                                                | 7                                          | 8                                         | 1                                          |
| Human water  | Small  | 9                       | -                                                | -                                          | -                                         | 9                                          |
|              | Medium | 9                       | -                                                | -                                          | -                                         | 9                                          |
|              | Large  | 7                       | -                                                | -                                          | -                                         | 7                                          |
| Pond water   | -      | 35                      | 19                                               | 11                                         | 5                                         | -                                          |

sulfisoxazole and 24.8% to tetracycline. Rao et al. (2011) isolated 253 *E. coli* from food and drinking water. Out of 50 STEC isolates, 24 (48%) showed resistance to penicillin, erythromycin, streptomycin, tetracycline, ampicillin, cephalexime, amoxicillin, ciprofloxacin, kanamycin, norfloxacin, ofloxacin, chloramphenicol, colistin, amikacin and gentamycin. Sood et al. (2015) reported that the majority of the *E. coli* and *K. pneumoniae* isolates from the drinking water in Ahmedabad (India) were sensitive to cotrimoxazole, quinolones, chloramphenicol, aminoglycosides, third-generation cephalosporins and tetracyclins. However, all the

*E. coli* isolated from drainage water were resistant to cotrimoxazole, third-generation cephalosporins and gentamycin. About 50% isolates from the drainage supplies showed resistance to quinolones and tetracyclins. In the present study, similar results of resistance pattern were obtained. Most of the *E. coli* isolates from slurry of all farms (small, medium, large) were resistant to oxytetracycline (59.3%) followed by enrofloxacin (56%) and least for sulphadiazine (23.5%), while maximum resistance was reported from isolates obtained from slurry samples of large farms. In milk samples, the highest resistance was found for oxytetracycline

**Table 5** Dairy farm-wise distribution of resistance against antibiotics in *K. pneumoniae* based on disc diffusion method

| Matrix       | Farm   | <i>K. pneumoniae</i> isolates | MDR (resistant to all antibiotics) (no of farms) | Resistant to two antibiotics (no of farms) | Resistant to one antibiotic (no of farms) | Sensitive to all antibiotics (no of farms) |
|--------------|--------|-------------------------------|--------------------------------------------------|--------------------------------------------|-------------------------------------------|--------------------------------------------|
| Milk         | Small  | 10                            | -                                                | 2                                          | 4                                         | 4                                          |
|              | Medium | 12                            | 1                                                | 3                                          | 4                                         | 4                                          |
|              | Large  | 8                             | 1                                                | 1                                          | 5                                         | 1                                          |
| Slurry       | Small  | 22                            | 3                                                | 8                                          | 6                                         | 5                                          |
|              | Medium | 25                            | 7                                                | 8                                          | 5                                         | 6                                          |
|              | Large  | 31                            | 13                                               | 8                                          | 4                                         | 6                                          |
| Animal water | Small  | 7                             | 1                                                | 1                                          | 4                                         | 1                                          |
|              | Medium | 10                            | -                                                | 5                                          | 4                                         | 1                                          |
|              | Large  | 14                            | 4                                                | 4                                          | 3                                         | 3                                          |
| Human water  | Small  | 3                             | -                                                | -                                          | -                                         | 3                                          |
|              | Medium | 2                             | -                                                | -                                          | -                                         | 2                                          |
|              | Large  | 4                             | -                                                | -                                          | -                                         | 4                                          |
| Pond Water   | ---    | 33                            | 21                                               | 10                                         | 2                                         | -                                          |



(61.5%). From large farm slurry samples out of 34 *E. coli* isolates, 16 were resistant to all three antibiotics (47.0%). Out of 31 *K. pneumoniae* isolates from slurry samples from large farm 13 (41.9%) were resistant to all three antibiotics.

*E. coli* isolates from dairy farm manure were found to be resistant to cephalothin, ampicillin, tetracycline and erythromycin in the USA (Walczak and Xu 2011). Cho et al. (2012) subjected 163 *E. coli* strains for susceptibility patterns to 16 antimicrobial agents and most of the tested isolates revealed higher antimicrobial resistance to ampicillin, cephalothin and gentamicin. In a study conducted by Skockova et al. (2012) in Czech Republic, of 89 samples of raw cow's milk, 84 (94.4%) were positive for *E. coli*. A total of 102 isolates were obtained and 50 isolates (49.0%) of these 102 isolates were observed resistant to tetracycline. A total of 233 *E. coli* isolates obtained from 175 tap water samples in Dhaka, Bangladesh (Talukdar et al. 2013). Nearly 36% ( $n = 84$ ) of the isolates were multi-drug resistant (MDR) and 26% ( $n = 22$ ) of these were positive for extended spectrum b-lactamase (ESBL).

### Prevalence of antibiotic-resistant genes in isolates from different matrices and farm types in Ludhiana

To assess the resistance genes in different matrices, genotypic profile was conducted for nine resistance genes. Three resistant genes each were targeted for oxytetracycline, sulphadiazine and enrofloxacin. For oxytetracycline, resistant genes *tetA*, *tetB* and *tetC* were targeted and isolates yielded expected amplicons at 502 bp, 173 bp and 888 bp (Fig. 3). Enrofloxacin-resistant genes *qnrA*, *qnrB* and *qnrC* from isolates yielded desired product at 516 bp, 469 bp and 417 bp (Fig. 4). While for sulphadiazine *sulI*, *sulII* and *sulIII* genes, *sul III* was not identified in any of the isolate and *sul I* and *sul II* PCR amplicon was at 433 bp and 293 bp (Fig. 5). Isolates which amplified the respective genes were sent for sequencing and samples were sequenced in both the directions. The NCBI BLAST software was used for analysing the sequence results

(Altschul et al. 1990) and was used as standard for study. The highest proportion of resistance genes was found in slurry (193) followed by milk (71). Overall pattern of resistant genes was *tetA* > *sulII* > *qnrS*. The large farms had shown the highest frequency of resistant genes both for *E. coli* and *K. pneumoniae*.

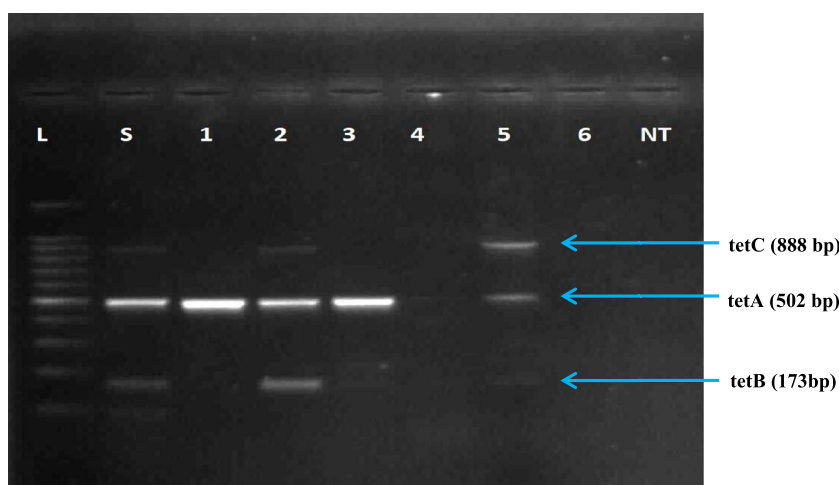
Overall relative frequency of resistant genes in *E. coli* isolates from different matrices and farm types in Ludhiana was maximum in slurry samples followed by milk and animal drinking water. No resistant gene was found in the isolates obtained from human drinking water (Table 6). In all matrices, the relative frequency of resistance genes in positive isolates of *E. coli* was as follows: *tetA* > *tetB* > *tetC* (53.5% > 24.7% > 0.59), *qnrS* > *qnrB* > *qnrA* (35.2% > 28.8% > 2.9%), *sulII* > *sulI* > *sulIII* (34.1% > 14.1% > 0%).

In the case of *K. pneumoniae* isolates, the overall relative frequency of resistant genes followed the same pattern as was for *E. coli*. The frequency of resistant genes was maximum in slurry samples followed by milk, animal drinking water (Table 6). No resistant gene was found in the isolates obtained from human drinking water. In all matrices, the relative frequency of resistance genes in positive isolates of *K. pneumoniae* was as follows: *tetA* > *tetB* > *tetC* (53.2% > 20.1% > 2.9%), *qnrS* > *qnrB* > *qnrA* (40.3% > 26.6% > 5.0%), *sulII* > *sulI* > *sulIII* (31.7% > 16.6% > 0%).

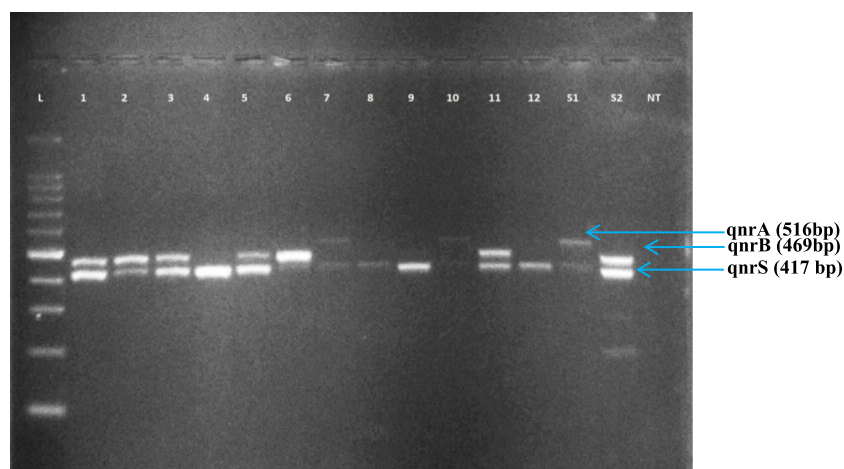
### Co-occurrence of antibiotic residues and resistant genes from different farms

Antibiotic residues, antibiotic-resistant bacteria (ARB) and antibiotic resistance genes (ARGs) are considered new classes of contaminants due to their potential adverse effects on human health. Information on the co-occurrences of the three important components of antibiotic resistance i.e. antibiotic residues (AB), ARB and ARGs in the same environmental samples is essential to understand that environment may play an important role in growing resistance.

**Fig. 3** Gel electrophoresis of PCR products of tetracycline resistant genes



**Fig. 4** Gel electrophoresis of PCR products of quinolone resistant genes



#### Co-occurrence of antibiotic residues and resistant genes of *E. coli* in milk from different farms

Concomitant presence of oxytetracycline residues in farms and resistant genes indicated that antibiotic residues leads selection pressure on organism leading them to gain resistance. Oxytetracycline residues were detected from one small farm, and simultaneously, *tetA* gene was noticed in that isolate. From two medium farms, oxytetracycline residues were detected and *tetA* and *tetB* gene were detected simultaneously. Similarly, in two large farms, *tetA* and *tetB* both were identified. Enrofloxacin residues were detected from three small farms, and simultaneously, *qnrS* and *qnrB* genes were prevalent in these farms respectively. From three medium farms, enrofloxacin residues were detected while *qnrB* and *qnrS* genes were detected simultaneously, while from two large farms, *qnrB* and *qnrS* both were identified respectively (Annexure. 1).

#### Co-occurrence of antibiotic residues and resistant genes of *K. pneumoniae* in various farms from milk

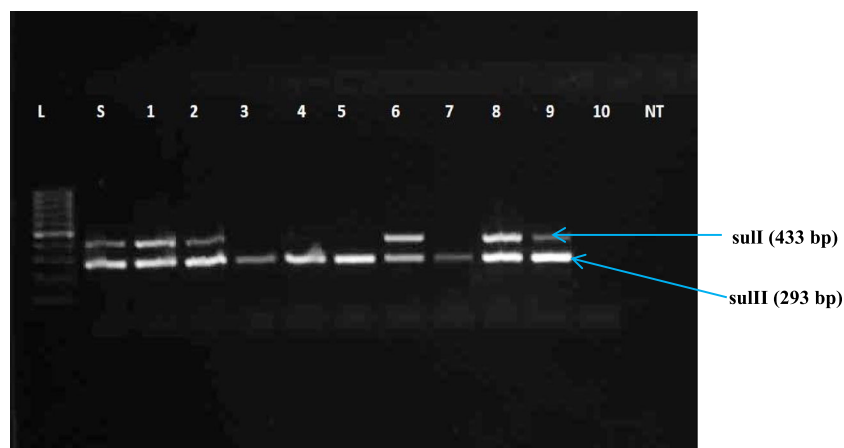
Oxytetracycline residues were detected from one medium farm and *tetA* and *tetB* were detected simultaneously in farm

isolate in milk sample. One large farm isolate possess *tetA* gene. Enrofloxacin residues were detected in milk from one small farm, and simultaneously, *qnrB* and *qnrS* genes were prevalent. From two medium farms, enrofloxacin residues were detected while *qnrS* gene was detected simultaneously in both farms. Enrofloxacin residues were found in two large farms and *qnrA*, *qnrB* and *qnrS* were identified simultaneously (Annexure. 2).

#### Co-occurrence of antibiotic residues and resistant genes of *E. coli* in various farms from slurry

Oxytetracycline residues were detected from 4 small farms, and simultaneously, *tetA* and *tetB* were detected. From two medium farms, *tetA* and *tetB* were detected simultaneously in farm isolates in slurry sample. While from 2 large farms, *tetA* and *tetB* were identified in both farm slurry samples. Enrofloxacin residues were detected from 2 small farms and simultaneously *qnrS* in both isolates. From 2 medium farms, *qnrB* and *qnrS* were detected simultaneously in farm isolates in slurry sample. While from the large farms, *qnrA*, *qnrB* and *qnrS* genes were all reported (Annexure. 3).

**Fig. 5** Gel electrophoresis of PCR products of sulphonamide resistant genes



**Table 6** Distribution of antibiotics resistance genes in farm milk, slurry and water

| Farm   | Matrix       | <i>E. coli</i> isolates | <i>tetA</i> | <i>tetB</i> | <i>tetC</i> | <i>sulI</i> | <i>sulII</i> | <i>sulIII</i> | <i>sulIII</i> | <i>qnrA</i> | <i>qnrB</i> | <i>qnrS</i> | <i>K. pneumoniae</i> isolates | <i>tetA</i> | <i>tetB</i> | <i>tetC</i> | <i>sulI</i> | <i>sulII</i> | <i>sulIII</i> | <i>qnrA</i> | <i>qnrB</i> | <i>qnrS</i> |
|--------|--------------|-------------------------|-------------|-------------|-------------|-------------|--------------|---------------|---------------|-------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|--------------|---------------|-------------|-------------|-------------|
| Small  | Milk         | 14                      | 5           | 3           | -           | 3           | 3            | -             | -             | -           | 4           | 4           | 10                            | 4           | -           | -           | 1           | 2            | -             | -           | 1           | 2           |
|        | Slurry       | 27                      | 13          | 3           | 1           | 4           | 8            | -             | -             | -           | 5           | 9           | 22                            | 11          | 3           | -           | 4           | 6            | -             | 1           | 5           | 8           |
|        | Animal water | 11                      | 4           | 1           | -           | -           | 2            | -             | -             | -           | 2           | 1           | 7                             | 4           | -           | -           | 1           | 1            | -             | -           | 3           | 1           |
|        | Human water  | 9                       | -           | -           | -           | -           | -            | -             | -             | -           | -           | -           | 3                             | -           | -           | -           | -           | -            | -             | -           | -           | -           |
| Medium | Milk         | 13                      | 8           | 5           | -           | 1           | 5            | -             | -             | -           | 6           | 4           | 12                            | 4           | 3           | -           | 1           | 2            | -             | -           | 2           | 6           |
|        | Slurry       | 30                      | 17          | 7           | -           | 6           | 11           | -             | -             | 2           | 10          | 11          | 25                            | 14          | 6           | -           | 5           | 9            | -             | 2           | 10          | 9           |
|        | Animal water | 13                      | 8           | 2           | -           | 1           | 2            | -             | -             | -           | 5           | 2           | 10                            | 7           | -           | -           | 2           | 1            | -             | -           | 1           | 3           |
|        | Human water  | 9                       | -           | -           | -           | -           | -            | -             | -             | -           | -           | -           | 2                             | -           | -           | -           | -           | -            | -             | -           | -           | -           |
| Large  | Milk         | 11                      | 6           | 2           | -           | 1           | 3            | -             | -             | -           | 4           | 5           | 8                             | 3           | 2           | -           | 1           | 2            | -             | 2           | 3           | 4           |
|        | Slurry       | 34                      | 19          | 16          | -           | 6           | 20           | -             | -             | 3           | 11          | 17          | 31                            | 19          | 11          | 4           | 5           | 17           | -             | 2           | 9           | 15          |
|        | Animal water | 17                      | 11          | 3           | -           | 2           | 5            | -             | -             | -           | 2           | 7           | 14                            | 8           | 3           | -           | 3           | 4            | -             | -           | 3           | 8           |
|        | Human water  | 7                       | -           | -           | -           | -           | -            | -             | -             | -           | -           | -           | 4                             | -           | -           | -           | -           | -            | -             | -           | -           | -           |

### Co-occurrence of antibiotic residues and resistant genes of *K. pneumoniae* in various farms from slurry

Oxytetracycline residues were detected from slurry samples of two small farms, and simultaneously, *tetA* gene was present in *K. pneumoniae* isolates of both farms. From 2 medium farms, two genes *tetA* and *tetB* were detected in both farm isolates, while from the large farm, *tetA* and *tetB* genes were reported. Enrofloxacin residues were detected from two small farms. In the first farm, *qnrB* and *qnrS* resistant genes were identified. While from other small farm, *qnrS* gene was present. From 3 medium farms, *qnrB* and *qnrS* were detected in slurry samples. While from large farms, all 3 resistant genes *qnrA*, *qnrB* and *qnrS* were reported (Annexure. 4).

The association of presence of antibiotic residues and resistance genes in all three types of dairy farms was examined statistically by chi-square test. The *p* values of < 0.05 prove that there is a statistically significant association between the presence of antibiotic residues and resistance genes (Table 7).

In India, Srinivasan et al. (2007) analysed 129 *E. coli* isolates from milk. Amongst 44 sulfisoxazole-resistant *E. coli* isolates, 1 isolate (2.3%) carried both *sulI* and *sulIII* genes. 10.9% *E. coli* isolates carried *tetA* gene, 10.1% carried *sulI* and 8.2% *sulIII*. Amongst 32 tetracycline-resistant isolates, 14 (43.8%) carried both *tetA* and *tetC* and 14 (43.8%) carried *tetC*. Rao et al. (2011) reported 253 isolates of *E. coli* from food and drinking water in Punjab, and the most common resistance determinants were *tetA* (60%) and *tetB* (27%). The results of the present study agree with previous studies of India. Similar genes of tetracycline and sulphonamide were detected in milk and slurry samples.

Wang et al. (2014) reported the frequency of sulphonamide resistance genes was *sulI* > *sul2* > *sul3* in pig-manured soil and *sul2* > *sul1* > *sul3* in chicken-manured soil in Jiangsu Province, China. Lyimo et al. (2016) conducted a study in northern Tanzania in drinking water sources in which amongst the tetracycline-resistant isolates, *tetA* was predominant (57.4 %) followed by *tetB* (24.0 %). In Romania, a study by Tabaran et al. (2016) on marketed raw milk and traditional dairy cheeses showed that 48.1% (*n* = 13) and 14.8% (*n* = 4) samples tested positive for genes encoding for tetracycline resistance *tetA* and sulphonamide resistance (*sulI*). Wang et al. (2016) detected 14 tetracycline resistance genes (TRGs) encoding ribosomal protection proteins (RPP) *tetM*, *tetO*, *tetQ*, *tetW*, *tetS*, *tetT* and *tetB/P*, efflux proteins (EFP) (*tetA*, *tetB*, *tetC*, *tetE*, *tetG*, *tetK*, *tetL* and *tetA/P*) and enzymatic inactivation proteins (*tetX*), three sulfonamide resistance genes (SRGs) (*sul1*, *sul2*, *sul3* and *sulA*) in animal wastewater in China.

**Table 7** Association of antibiotic residues and resistance genes

|                        | Antibiotic residues | Antibiotic resistance | Chi-square | p value |
|------------------------|---------------------|-----------------------|------------|---------|
| Milk                   |                     |                       |            |         |
| Small dairy farms (51) | 5                   | 11                    | 6.1        | 0.03    |
| Medium farms (51)      | 9                   | 12                    | 3.7        | 0.05    |
| Large farms (51)       | 4                   | 10                    | 17.8       | 0.00    |
| Slurry                 |                     |                       |            |         |
| Small dairy farms (51) | 9                   | 22                    | 7.6        | 0.01    |
| Medium farms (51)      | 9                   | 26                    | 0.45       | 0.52    |
| Large farms (51)       | 5                   | 33                    | 0.56       | 0.45    |

In the current study, in all matrices, the relative frequency of resistance genes in positive isolates of *E. coli* was as follows: *tetA* > *tetB* > *tetC* (53.5% > 24.7% > 0.59), *qnrS* > *qnrB* > *qnrA* (35.2% > 28.8% > 2.9%), *sulII* > *sulI* > *sulIII* (34.1% > 14.1% > 0%). While in all matrices, the relative frequency of resistance genes in positive isolates of *K. pneumoniae* was as follows: *tetA* > *tetB* > *tetC* (53.2% > 20.1% > 2.9), *qnrS* > *qnrB* > *qnrA* (40.3% > 26.6% > 5.0%), *sulII* > *sulI* > *sulIII* (31.7% > 16.6% > 0%). The highest proportion of resistance genes was found in slurry (193 genes) followed by milk (71 genes). Overall pattern of resistant genes was *tetA* > *sulII* > *qnrS*. The large farms had shown the highest frequency of resistant genes both for *E. coli* and *K. pneumoniae*.

## Conclusions

In conclusion, data from the present study suggested that commensal *E. coli* and *Klebsiella* from the study farms may act as reservoirs of antimicrobial drug resistance genes which may be mobilised into human populations. To the best of our knowledge, this is the first comprehensive study of molecular mechanisms of resistance in commensal *E. coli* and *Klebsiella* associated with livestock farms in Ludhiana, Punjab. Results should however be interpreted with caution as a limited number of isolates were investigated and a comprehensive history of antibiotic use necessary to make informed decision on the contribution of antibiotic use to the observed resistance is not available. It however pointed attention to the critical need for the regulation of antimicrobial drug usage in livestock production and continuous monitoring of antibiotic resistance in developing countries for public health safety. The study also concludes that animal farm waste (slurry) may consider an indicator to assess the antimicrobial usage and subsequently their release into environment. 32.9 % of *E. coli* and 29.4 % *Klebsiella* isolates were resistant to

all 3 antibiotics in slurry and 21% of *E. coli* and 6.7% of *Klebsiella* isolates in milk were resistant to all 3 antibiotics. The genotypic expression of the relative frequency of resistance genes in positive isolates of *E. coli* and *Klebsiella* was as follows: *tetA* > *tetB* > *tetC* for tetracyclines, *qnrS* > *qnrB* > *qnrA* for enrofloxacin, *sulII* > *sulI* > *sulIII* for sulphonamides. The highest proportion of resistance genes was found in slurry (193 genes) followed by milk (71 genes) and the overall pattern of resistant genes was *tetA* > *sulII* > *qnrS*. Untreated animal waste may be considered an important source of resistant bacteria leading to environmental pollution.

**Authors' contributions** All authors contributed to the study conception and design. Material preparation, data collection, investigation, analysis and validation were performed by Prateek Jindal and Jasbir Bedi. Resources were provided by R.S. Aulakh and J.P.S. Gill. The first draft of the manuscript was written by Prateek Jindal. Supervision was done by Jasbir Bedi, Randhir Singh and R.S. Aulakh. Funding acquisition and project administration were done by Jasbir Bedi and J.P.S. Gill. All authors helped in the review and editing of the manuscript. All authors read and approved the final manuscript.

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## Compliance with ethical standards

**Competing interests** The authors declare that they have no competing interests.

**Ethical approval** Ethical approval was obtained by institutional animal ethics committee (IAEC), Guru Angad Dev Veterinary and Animal Sciences University, Ludhiana (Registration no. 497/GO/Re/SL/01/CPCSEA). The ethical approval number allotted to the current study on consent was GADVASU/2017/IAEC/41/13.

**Consent to participate** As part of the informed consent process, the consent document was designed to provide information to potential subjects about a research study so they can make an informed decision about their participation.

**Consent to publish** Not applicable

## Annexure 1

**Table 8** Concomitant presence of antibiotic residues and resistance genes in *E. coli* in farm milk

| Oxytetracycline residue positive | <i>tet A</i> | <i>tetB</i> | <i>tetC</i> |
|----------------------------------|--------------|-------------|-------------|
| Small                            | +            | -           | -           |
| Medium                           | +            | +           | -           |
|                                  | -            | +           | -           |
| Large                            | +            | -           | -           |
|                                  | +            | +           | -           |
| Enrofloxacin residue positive    | <i>qnrB</i>  | <i>qnrS</i> | <i>qnrA</i> |
| Small                            | -            | +           | -           |
|                                  | +            | -           | -           |
|                                  | +            | -           | -           |
| Medium                           | +            | -           | -           |
|                                  | -            | +           | -           |
|                                  | +            | +           | -           |
| Large                            | +            | +           | -           |
|                                  | -            | +           | -           |

## Annexure 2

**Table 9** Concomitant presence of antibiotic residues and resistance genes in *K. pneumoniae* in farm milk

| Oxytetracycline residue positive | <i>tet A</i> | <i>tetB</i> | <i>tetC</i> |
|----------------------------------|--------------|-------------|-------------|
| Medium                           | +            | +           | -           |
| Large                            | +            | -           | -           |
| Enrofloxacin residue positive    | <i>qnrA</i>  | <i>qnrB</i> | <i>qnrS</i> |
| Small                            | -            | +           | +           |
| Medium                           | -            | -           | +           |
|                                  | -            | -           | +           |
| Large                            | +            | +           | +           |
|                                  | -            | -           | +           |

## Annexure 3

**Table 10** Concomitant presence of Oxytetracycline residues and resistance genes in *E. coli* in farm slurry

| Oxytetracycline residue positive | <i>tet A</i> | <i>tetB</i> | <i>tetC</i> |
|----------------------------------|--------------|-------------|-------------|
| Small                            | +            | +           | -           |
|                                  | +            | -           | -           |
|                                  | +            | -           | -           |
|                                  | -            | +           | -           |
| Medium                           | +            | -           | -           |
|                                  | +            | +           | -           |
| Large                            | +            | +           | -           |
|                                  | +            | +           | -           |
| Enrofloxacin residue positive    | <i>qnrA</i>  | <i>qnrB</i> | <i>qnrS</i> |
| Small                            | -            | -           | +           |
|                                  | -            | -           | +           |
| Medium                           | -            | +           | -           |
|                                  | -            | -           | +           |
| Large                            | +            | +           | +           |
|                                  | -            | +           | +           |

## Annexure 4

**Table 11** Concomitant presence of antibiotic residues and resistance genes in *K. pneumoniae* in farm slurry

| Oxytetracycline residue positive | <i>tet A</i> | <i>tetB</i> | <i>tetC</i> |
|----------------------------------|--------------|-------------|-------------|
| Small                            | +            | -           | -           |
|                                  | +            | -           | -           |
| Medium                           | +            | +           | -           |
|                                  | +            | +           | -           |
| Large                            | +            | +           | -           |
| Enrofloxacin residue positive    | <i>qnrA</i>  | <i>qnrB</i> | <i>qnrS</i> |
| Small                            | -            | +           | +           |
|                                  | -            | -           | +           |
| Medium                           | -            | +           | -           |
|                                  | -            | +           | +           |
|                                  | -            | -           | +           |
| Large                            | +            | +           | +           |
|                                  | -            | +           | +           |



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